

Review retrospective:
*the anharmonic FDT identities, and a docstring written ahead
of the code*

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Abstract

A soundness review of `Laplace/OneD/AnharmonicFDT.lean`: the two conditional fluctuation–dissipation / three-point identities for the anharmonic Gibbs measure — $\partial_h \langle x \rangle_h|_0 = -t \text{Var}_0(x)$ and $\partial_h \text{Cov}_h(x, x)|_0 = -t \kappa_3(x, x, x)$. Result: **a clean fidelity pass with five edits** — two anticipatory dead imports and three docstring over-claims. The compositions are correct; the docstring had simply been written ahead of the code.

1 Setting

The anharmonic analogue of the harmonic cross-susceptibility capstone: the same Threepoint abstract identities (`gibbsExp_deriv_eq_neg_t_cov`, `gibbsCov_deriv_eq_neg_t_kappa3`), specialised at $A = B = \varphi = \text{id}$ to the anharmonic potential and made unconditional in the observability side conditions by the anharmonic monomial witnesses. Where the harmonic κ_3 vanished by parity, here the cubic term keeps it.

2 What was reviewed

Two theorems: the first-cumulant FDT and the cross-susceptibility three-point identity.

3 Fidelity / soundness findings

The compositions are sound. The FDT specialises `gibbsExp_deriv_eq_neg_t_cov` with the regularity witness and observable witnesses for $\varphi = x$ and $\varphi \cdot A = x^2$; the three-point identity fills the six `GibbsObservable` slots of `gibbsCov_deriv_eq_neg_t_kappa3` as `hx hx hxx hxx hxx hxxx` (two x , three x^2 , one x^3 , all coinciding under $A = B = \varphi = \text{id}$). The reviewer confirmed the multiplicities; no witness mismatch.

Three docstring over-claims, fixed. (1) The `## Headlines` section advertised four headlines, but only two exist as theorems — the two `*_asymptotic` capstones are an unwritten TODO. Split into “present in this file” and “Planned (TODO)”. (2) “the two headline identities are therefore **unconditional**” was too strong: they discharge the `GibbsObservable` side conditions but still require $\lambda, \gamma > 0$, $\alpha^2 < 3\lambda\gamma$, $t > 0$. Qualified. (3) “the cubic term breaks the parity argument” over-claims, since the theorem does not assume $\alpha \neq 0$ — at $\alpha = 0$ the potential is even and κ_3 may still vanish. Softened to “no parity simplification is built in”.

4 Simplifications applied

Two dead imports. `Laplace.OneD.AnharmonicKappa3` and `Laplace.OneD.IntegralRemainder` are referenced by no symbol — only named in the TODO prose for the unwritten asymptotic capstones. A full lake build (8306 jobs) stayed green with both removed, confirming they are not transitively load-bearing. `open MeasureTheory` (bare `Measure  $\mathbb{R}$ /volume`) and the three used imports stay. Signatures byte-identical, no warnings. GPT affirmed.

5 Disagreements and open questions

None. The reviewer’s trailing-`exact` removal was declined: the proofs are tactic blocks with preceding `haves`, so the closing `exact <application>` is not cleanly convertible to a term — proof-golf, out of scope.

6 Lessons

- **A “Headlines” list can run ahead of the declarations.** The module doc advertised two `*_asymptotic` theorems that are still a TODO — a reader scanning the headlines would expect them to exist. When a file documents planned work inline, keep the headline list to what actually type-checks, and quarantine the rest under a “Planned” heading.
- **Anticipatory imports are dead imports until the anticipated code lands.** The two imports were placed for capstones not yet written; until then they are removable dead code (the future tide re-imports as it needs). The build, not the intent, decides — and here the build confirmed them inert.
- **“Unconditional” needs a scope.** The identities are unconditional *in the observability side conditions*, not in the model parameters; an unqualified “unconditional” misleads about λ, γ, t and the discriminant.

7 Follow-ups

- The asymptotic capstones the TODO describes (`gibbsExp_deriv_anharmonic_asymptotic  $\rightarrow -1/\lambda$` , `gibbsCov_deriv_anharmonic_asymptotic  $\rightarrow \alpha/\lambda^3$`) — a forward tide that will re-add the two imports. This review supplies the “GPT confirms the conditional forms are correct” precondition the TODO asked for.
- Remaining `OneD/Anharmonic*` `arc` (`AnharmonicGibbsRegularity`, `AnharmonicKappa3Affine`, `AnharmonicFourthCumulant`, ...) and `IntegralRemainder` (2668L).